Planar Keller Maps after the Counterexample: Machine-Certified Newton-Polygon Exclusions, the Staircase Transport, and the Recalibrated Frontier

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Abstract

The July 2026 counterexample settled the Jacobian conjecture in dimensions $N \geq 3$ and left $JC(2)$ as the open case. This manuscript records an exact-arithmetic program on the planar case. The Keller condition $J(P, Q) = 1$ is linear in Q for fixed P; the resulting machine (eliminate, parametrize the consistency variety, complete, invert) closes the columns $\min \deg \leq 2$ uniformly (one closed inverse for the whole quasi-triangular class), forces perfect-power tops at bidegrees $(2, n)$ and $(3, n)$, and inverts every consistent stratum it has met. Four unconditional exclusion theorems follow from a weight-class calculus: a two-term $P = x + ax^u y^v$ is never a Keller component at any partner degree; neither is any lower-weight perturbation of it; no Keller component carries a Newton top edge anchored at the linear vertex x with interior direction; and if x is a vertex of the Newton polygon then $P = x + f(y)$ exactly (the vertex dichotomy). The residual open core is the transport of the Keller constant along the staircase of Newton vertices swallowing x: we show the completion window is block-triangular under the edge grading, exhibit the classwise transport chain as an executable elimination whose obstruction always localizes at the constant's class, and prove the corresponding fifth exclusion: for primitive tops the certificate-tower induction closes it UNCONDITIONALLY at all partner degrees (cleared base certificates with monomial pairings such as $-13860\,a^7$, plus a resonance-reduction lemma showing window growth can never repair the constant's row). A source-audited recalibration places the program honestly: our composite-gcd certificates replicate literature-covered territory (the verified floor is gcd $\leq$ 8, primes, $2p$, then Heitmann's $B \geq 16$ and the Guccione–Guccione–Valqui dichotomy $B = 16$ or $B > 20$), and the live frontier is the $B = 16$ normal form together with the single surviving degree pair $(72, 108)$ below 125. For that open pair, an exact primary-source instantiation gives two original-pair rejection gates: seven possible northeastern vertices for a nonzero inner polynomial, and $\deg_x \operatorname{Res}_y(P, Q) = 21$ with an $84 + 24$ major/minor root partition; neither gate alone excludes the pair. A constructible determinantal-strata calculation now closes the first actual three-coefficient GGHV slice exactly. On the residual curve, two normalized maximal minors reduce, with $z = u^9$, to $21 - 96z - 1024z^2$ and $(8z + 1)^{14}$; a persisted integer Bezout identity has constant 17^{14}. After adding $\varepsilon_{(0,7)}$, the only two-chart residual is a degree-12 fiber over $z = -1/8$. A third exact minor has coprime degree-13 fiber determinant, certified by a second integer Bezout identity. A connected mod-9 row/column grading extends to every nonconstant direction in the persisted lower family. This is a strict slice result and does not exclude the full $(72, 108)$ family. The full-family audit then removes the structural constant column, corrects the complete row count from 289 to 302, and compresses one selected augmented minor to a 36-column block depending on 24 of the 51 parameters. Two connectivity-generating triples factor exactly; one graph-active direction cancels identically, while the other determinant becomes weighted homogeneous with weights $(7, 3, 9)$ after shifting the forced coefficient. On the $d = 0$ boundary, the tautological kernel $[P, P] = 0$ can be quotiented exactly. One quotient determinant factors into b^{32}, one five-term weighted factor, and five binomials; a second is a nonzero scalar times a^{107}. Together with the exact origin rank gap, these charts close the complete quotient boundary. On $d \neq 0$, a first alternative chart reduces each selected component G, L, Q to a finite intersection. A third exact determinant factors as $X^{30}LQR$, for $X = A^3$, and exact component ideals close G. The L and Q common residuals remain finite. These are strict chart results, not closure of the full T_B restriction or the full family. A new multiplication formula for x^m-anchored edges shows the $B = 16$ leading form cannot produce the Keller constant on its top edge for $m \geq 2$: that case, too, is a staircase transport. All computations are exact and reproducible; every claim is labeled machine-verified [MV], derived [D], or conjectural [C].

1 Introduction

Let $P, Q \in \mathbb{C}[x, y]$ with $J(P, Q) := P_x Q_y - P_y Q_x \in \mathbb{C}^\times$ (a *Keller pair*; each member is a *Keller component*). The two-variable Jacobian conjecture $JC(2)$ asserts every Keller pair is an automorphism pair. Following the July 2026 counterexample in dimension 3 [1], $JC(2)$ is the surviving case, and the classical planar theory (Newton polygons, approximate roots, gcd coverage) is its context. This manuscript is the planar half of a two-manuscript record: the foundational companion documents the three-dimensional aftermath (validation, the seed family, the geometry of non-properness, two-dimensional equivariant rigidity); this one documents the planar program. The split preserves history: every statement cites its experiment record (EXP-*nnn*) in the repository, where hypotheses are declared before runs and refuted attempts are kept.

Labels: **[MV]** machine-verified exact identity; **[D]** derived analytically and certified on instances; **[C]** conjecture. Machine-verification is exact sympy arithmetic over $\mathbb{Q}$ or $\mathbb{Q}(i)$; no floats anywhere.

Two structural inputs from the companion are used freely. First, every $\mathbb{G}_m$-equivariant Keller map of $\mathbb{C}^2$ is an automorphism; on the opposite-sign/one-zero scope relevant to the three-dimensional mechanism it is linear (EXP-010; independently classified for all signatures by Shaska [18]). Second, at every weight ray the leading pair of a planar Keller map is Jacobian-dependent (the tropical bridge, EXP-013), which is the entry point of all edge analyses below.

2 The planar machine

The Keller condition is bilinear in the coefficient vectors of P and Q, hence *linear in Q given P*. In the affine gauge (identity linear parts) this yields a machine per degree pair: eliminate Q (the consistency ideal in P's coefficients), parametrize, complete linearly, and invert or fiber-test (EXP-017, EXP-019, EXP-020).

Theorem 2.1 ([MV]; uniform closure at $\min\deg \le 2$ (EXP-021/022))**.** *Write $\ell = \alpha x + \beta y$, $\beta \neq 0$. In shear coordinates the Keller condition for $P = x + f(\ell)$ (any polynomial f) is a linear first-order PDE with characteristic invariant P; its complete polynomial solution space is $Q = \ell/\beta + H(P)$, H arbitrary, and the pair inverts by one closed formula. Consequently every planar Keller map with $\min(\deg P, \deg Q) \le 2$ is invertible, and the whole quasi-triangular class $P \sim x + f(\ell)$ is invertible at every degree.*

Theorem 2.2 ([MV]; forced tops at $(2, n)$ and $(3, n)$ (EXP-019/020/022/039))**.** *The $(2, 3)$ consistency ideal is the single equation $(4A_0A_2 - A_1^2)^2 = 0$ (P's top form is a perfect square); the $(2, 4)$ elimination returns the same ideal. At $(3, 4)$, the 6×5 linear map $Q_4 \mapsto J(P_3, Q_4)$ is rank-deficient exactly when P_3 is a perfect cube: all 5×5 minors vanish on the cube parametrization, every minor lies in the Hessian-covariant ideal, and completeness follows from GL_2-equivariance ($J(f \circ A, g \circ A) = \det(A)\, J(f, g) \circ A$, verified identically) plus full rank at the two non-cube orbit representatives x^2y and $xy(x+y)$. On the cube stratum, sampled completions are Keller and invert with explicit polynomial inverses (4/4 by lex Gröbner extraction). At $(3, 3)$ the consistency ideal forces the quasi-triangular alignment at the radical level.*

The degree columns close by classical coverage: $\gcd(2, n) \le 2$ and $\gcd(3, n) \in \{1, 3\}$ are inside the verified gcd landscape of Section 5, so the machine's contribution there is the *constructive* loop (explicit consistency varieties and inverses), stated as replication. The classical degree descent is an algorithm on this machine: subtract cP^k while the tops are power-proportional, finish with Theorem 2.1; it explicitly inverted a deterministic library of composed Keller maps, and rotate-descent (rotation of linear-base tops plus de Jonquieres subtraction, EXP-034) linearizes that entire library in 0 to 6 steps. Its exact limit: mixed-corner tops stay mixed under rotation; those are the classical hard shapes, and they are where the exclusion theorems below take over.

3 Four unconditional exclusions

Throughout, P has linear part x after affine gauge. The weight-class calculus: under $w = (v, 1-u)$, the operator $J(P, \cdot)$ shifts weight by a constant, so the Keller equation decouples into weight classes; the technique goes back to Dixmier and Joseph [14] and is standard in the polygon literature [8]; the statements below appear to be new in the stated generality (Section 6).

Theorem 3.1 ([MV/D]; the weight-class theorem (EXP-029))**.** *For $u \ge 2$, $v \ge 1$, $a \neq 0$, the polynomial $P = x + a\, x^u y^v$ is not a Keller component: no Q, of any degree, satisfies $J(P, Q) \in \mathbb{C}^\times$.*

Proof shape. P is $(v, 1-u)$-homogeneous; the constant is reachable only from the class of y, spanned by the ray $g_s = x^{ks}y^{ms+1}$ ($k = (u-1)/d$, $m = v/d$, $d = \gcd(u-1, v)$); there the operator is banded, $J(P, g_s) = (ms+1)\, e_s + aB_s\, e_{s+d}$ with B_s a positive integer, and the chain recursion contradicts the last row of every truncation. □

Theorem 3.2 ([MV/D]; lower-weight perturbations never rescue (EXP-030/031))**.** *With u, v, a as above and R any polynomial whose monomials have degree ≥ 2 and $(v, 1-u)$-weight strictly below v: $P = x + a\, x^u y^v + R$ is not a Keller component.*

Theorem 3.3 ([MV/D]; every x-anchored edge falls (EXP-032))**.** *Let $z = x^k y^m$, $k, m \ge 1$, and $E = x\,\varphi(z)$, $\deg\varphi = g \ge 1$. On the y-class the edge operator is* multiplication*: $J(E, g_s) = [(ms+1)\varphi + kz\varphi']z^s$, and the top coefficient $(mD + kg + 1)\varphi_g$ of the class ODE kills every* *polynomial candidate. Hence E is never the leading form of a Keller component, and $P = E + R$ (any strictly lower-weight tail) is never a Keller component.*

Theorem 3.4 ([MV/D]; the vertex dichotomy (EXP-033))**.** *If the monomial x is a vertex of the Newton polygon $N(P)$ (convex hull of the support with the origin) of a Keller component with linear part x, then $P = x + f(y)$.*

Lemma 3.5 (Annihilation (EXP-036))**.** *For any polynomial P, $J(m, P^k) = -k\,L(P^{k-1}m)$ with $L = J(P, \cdot)$ (Leibniz). Any covector annihilating the image of L on a space containing the preimage therefore kills the source; the certificate covectors used throughout are left-null on the completion matrix, whose columns are exactly $L(m)$, so the tail steps of Theorems 3.2–3.4 are unconditional. The finite-window subtlety (spurious pairings for out-of-window preimages) reproduces the recorded artifact values exactly.*

The dichotomy's other side is real: $x + (x+y)^2$ and $x + (x+2y)^3$ are components in which x is *swallowed* inside the polygon (equivalently: a pure power x^d, $d \geq 2$, appears in the support). Every non-triangular component is of that kind. The residual open core is therefore precise: components whose polygon swallows x with a mixed staircase.

4 The corner is diagonal; the staircase transport

Proposition 4.1 ([MV]; the diagonal corner (EXP-035))**.** *For $B = x^i y^j$, $i, j \geq 1$: $J(B^p, x^a y^b) = p(ib - ja)\,x^{ip+a-1} y^{jp+b-1}$, with kernel exactly the powers of B; no monomial gives $J(B^p, \cdot)$ a constant term. The mixed corner is clean; the obstruction lives on the staircase of Newton vertices joining the swallowed linear vertex to the corner.*

Theorem 4.2 ([MV]; block triangularity and the transport chain (EXP-037))**.** *Fix the x-edge grading $w = (v, 1-u)$ and let s_0 be the minimal w-class of P. In the completion window (unknown Q of bounded degree), every nonzero matrix entry sits at a class offset $s - \sigma$ for a P-class s ($\sigma = v + 1 - u$), and the offset $s_0 - \sigma$ is minimal: ordering Q-classes by w-class, the system is block triangular. Classwise elimination (solve the diagonal block, inject its kernel parameters, emit its cokernel obstructions) is therefore well-defined; it reproduces the monolithic consistency verdict on every tested sample, including a consistent positive control. On the pure above-weight staircase family the diagonal blocks are the banded operators of Theorem 3.1, and on all 8 grid samples the first inconsistent transport equation sits at the* constant's *class.*

Theorem 4.3 ([MV]; the fifth exclusion, window form (EXP-042))**.** *Let $P = x + a\,x^u y^v + b\,x^d$ ($u \geq 2$, $v, d \geq 1$; the minimal shape swallowing x). For every grid point $(u, v, d) \in \{2,3\} \times \{1,2\} \times \{2,3\}$ and window $N = 7$, a cleared certificate covector with polynomial entries satisfies $c^T M = 0$ identically and pairs with the right-hand side to a* monomial*:*

$$-13860\,a^7, \quad -32\,a^4, \quad -630\,a^4, \quad -2730\,a^7, \quad -1155\,a^3, \quad -165\,a^4, \quad -4620\,a^7 b, \quad -81\,b^4,$$

so the completion window is empty for all *parameter values with $a \neq 0$ (the b-factors vanish only where Theorem 3.1 applies anyway). At $(u, v, d) = (2, 1, 2)$ the pairing at window N is $-c_N\,a^N$ with $c_N \in \{420, 1260, 13860, 180180, \dots\}$ for $N = 5..9$: a nonzero monomial in a alone at every certified window. The annihilation mechanism of Lemma 3.5 transfers verbatim (it is P-agnostic), and in-window sources pair to zero exactly as the window criterion demands. A three-parameter tail certificate ($-27720\,a^7 c_3^4$) extends the family.*

Lemma 4.4 ([MV/D]; the tower lemma (EXP-044/046))**.** *Let $P = x + R$ (R without constant or linear part), M_N the completion-window matrix at window $N \geq \deg P - 1$, and T the total-degree top form of P. Suppose every form in $\ker J(T, \cdot)$ at each degree is a scalar multiple of a power that reduces, modulo $\mathbb{C}[P]$, to degree $\leq N$ (this holds whenever T is a primitive monomial, and whenever $T = P - x$ or T is the pure power bx^d, by expanding $T^k = \lambda P^k +$ lower). Then any left-null certificate of M_N with nonzero pairing extends to M_{N+1} with the same (scaled) pairing: window inconsistency propagates to every larger window. Proof ingredients, each machine-verified: old columns vanish on new rows (degree bookkeeping); the new block is $J(T, \cdot)$ on degree-$(N+1)$ forms with the predicted kernel; each resonance κ satisfies $\kappa - \lambda P^k \in$ window with $P^k \in \ker L$, so $L(\kappa)$ is an in-window image which every certificate annihilates; the extension was realized explicitly across an active resonance (window $8 \to 9$, ratio 2, pairing preserved). The measured window law ($-c_N a^N$ pairings with the odd-prime $2N - 3$ ratios) is this tower seen through the clearing normalization.*

Theorem 4.5 ([MV/D]; all-degree fifth exclusion (EXP-042/044/046))**.** *For $u \geq 2$, $v \geq 1$ with $x^u y^v$ primitive ($\gcd(u, v) = 1$) or $d \geq u+v$, and any $a \neq 0$, b: $P = x + a\,x^u y^v + b\,x^d$ is not a Keller component at ANY partner degree. (Base: the cleared certificates of Theorem 4.3; induction: Lemma 4.4.) For proper-power tops (e.g. $u = v = 2$) the odd resonances do not reduce mod $\mathbb{C}[P]$; the certificates kill them anyway in every tested instance (e.g. $(xy)^5$ at window 9), so the statement there is [D] pending the derivation of that kill.*

Theorem 4.6 ([MV/D]; the universal tower (EXP-049))**.** *Let $P = x + R$ (R without constant or linear part) whose total-degree top form $T = P_D$ is NOT a proper power. If the completion window is inconsistent with a certificate at any single $N \geq \deg P - 1$, then P is not a Keller component at ANY partner degree. (The tower argument with T the full top form: $\ker J(T, \cdot)$ on degree-M forms is zero unless $\deg T \mid M$ and then the span of $T^{M/\deg T}$; and $T = P -$ lower by definition, so $T^k - P^k$ has degree $\leq k \deg P - 1$: every resonance reduces in-window modulo $\ker L$.) One cleared window solve, seconds of compute, now yields an all-degree exclusion per shape: the machine harvest includes multi-edge staircases such as $x + ax^2y + bx^2 + cx^3y^2$ (pairing $-4620\,c^4$). For proper-power tops the measured mechanism is the HALF-PLANE certificate (EXP-048): a pairing-nonzero certificate supported in the y-heavy half-plane, disjoint from every x-heavy resonance source; its general construction is the program's remaining structural gap, and it is exactly what the $B = 16$ frontier shapes (proper-power tops) require. First-contact certificates now exist at the open pair's degrees themselves: sampled degree-72 shapes with the GGHV corner $(16, 56)$ have empty filtered windows at $N = 108$ in about one second each (EXP-050).*

Theorem 4.7 ([MV/D]; the half-plane tower (EXP-051))**.** *Let $P = x + R$ whose total-degree top form attains $\min\{p - q\}$ over the support (the swallowed-staircase geometry), and let H be the y-heavy half-plane of rows $\{i \leq j\}$. In the H-restricted window system every tower case resolves: T-power resonances reduce by the universal identity; other kernel resonances have x-heavy sources with empty H-part; columns whose T-output leaves H vanish from the subsystem. Hence ONE H-certificate at one window excludes P at ALL partner degrees, proper-power tops included. Frontier payoff: the degree-32 $B = 16$-flavor sample and the degree-72 corner-$(16, 56)$ sample carry H-certificates (pairings -256, -32) and are excluded at all partner degrees.*

Two instrument-level results complete the toolkit. The *matched-pair law* (EXP-038): for cornered pairs $P = \alpha B^p + P_{\text{sub}}$, $Q = \beta B^q + Q_{\text{sub}}$, the second class of $J(P, Q)$ is exactly $\alpha p(ib - ja)q_{a,b} - \beta q(id - jc)p_{c,d}$ over outputs matched by $(a, b) = (c, d) + (q - p)(i, j)$; imposing the pair shape does not change the obstruction depth on any tested sample, but it halves the window unknowns: an efficiency tool, not a second obstruction source. And the x^m*-anchored edge operator* (EXP-043): for $E = x^m\varphi(z)$, $z = x^k y^l$,

$$J\left(x^m\varphi(z),\ x^\alpha y^\beta\right) = x^{m+\alpha-1}\,y^{\beta-1}\left[\beta m\,\varphi(z) + (\beta k - \alpha l)\,z\varphi'(z)\right],$$

verified identically in symbolic edge coefficients; Theorem 3.3's operator is the case $m = 1$.

5 The frontier, recalibrated from primary sources

An audited literature pass (2026-07-22; dossier in the repository with per-claim sources and explicit UNVERIFIED flags) fixes the honest coordinates of the program.

Verified gcd coverage. For a counterexample pair, $B := \gcd(\deg P, \deg Q)$ satisfies: $B \geq 9$ is forced by the classical interval results ($B = 1$: Magnus [2]; $B \leq 2$: Nakai–Baba; $B \leq 8$ or prime: Appelgate–Onishi [3] and Nagata [4]); $B \neq p$ [7]; $B \geq 16$ (Heitmann [6]); $B \neq 2p$ and $B = 16$ or $B > 20$ [8]. Zoladek's claimed $B > 33$ has a documented gap [8] and is not used. Consequently our machine certificates at gcd $= 2, 9, 12, 18$ (EXP-024..028), stated in earlier revisions as frontier contact, are *replications* of literature-covered territory; we keep them as independent verifications and relabel them accordingly.

Verified degree floor. Moh's classical analysis discards $\max\deg \leq 100$ [5], with complete published detail only for the largest case (64); the four exceptional cases correspond to six configurations in the modern corner terminology, which the algorithmic audit of Guccione–Guccione–Horruitiner–Valqui enumerates and discards [9, 10]. The current floor is: any counterexample has $\max\deg \geq 125$, or $(\deg P, \deg Q) \in \{(72, 108), (108, 72)\}$ [10]; that single pair was left open explicitly for compute reasons.

The live targets. (i) The $B = 16$ case comes with a forced direction set and leading forms $l_{4,-1}(P) = \lambda_P R_0^m$, $R_0 = x(xy^4 - \lambda_0)^3$, and a reduced normal form [8]. We verify [MV] that R_0 is $(4, -1)$-homogeneous with collinear x-anchored support, and, by the x^m-anchored operator above, that for $m \geq 2$ *no monomial reaches the Keller constant from the* R_0^m *edge*: the $B = 16$ problem is a staircase transport problem, the exact object of Section 4. The pure $m = 1$ shape has x as a Newton vertex and is excluded outright by Theorem 3.4 (window replication empty); swallowed variants land in Theorem 4.3's territory (sampled windows empty). (ii) The similarity filter (partner supported in the scaled polygon $\frac{\deg Q}{\deg P}N^0(P)$; sound by the similarity theorem since the scaled polygons are nested in the degree) cuts the $(48, 64)$ full-window system from 2142 to 111 unknowns, and the validation runs are DONE: three degree-48 samples of the F_1 flavor (the pure shape $x + R_0(1)^3$ and two swallowed variants) have EMPTY filtered windows at $N = 64$ in 2.4 to 3.6 seconds each, excluding all partners of degree ≤ 64 per sample (the classical case-64 degrees: a sampled-level replication of Moh/Heitmann demonstrating the pipeline's cost at target scale). (iii) For the $(72, 108)$ system the histograms give 5992 unknowns in 535 classes with largest diagonal block 22 before filtering: with the filter and the transport, the open territory (max degree > 150, and $(72, 108)$ itself) is affordable at seconds-to-minutes per certificate.

The $(72, 108)$ attack: the reduced system and the perturbative covector

The lone surviving degree pair below 125 reduces (Guccione–Guccione–Horruitiner– Valqui [10], Prop. 4.3; transcription verified against the LaTeX sources) to the existence of a pair with

$[P, Q] = x^2$ supported in SMALL polygons:

$$\begin{aligned} N(P):\ & (0,0),(1,0),(8,14),(8,16),(0,8) && \text{(61 lattice points)}, \\ N(Q):\ & (0,0),(2,1),(12,21),(12,24),(0,12) && \text{(125 lattice points)}. \end{aligned}$$

The original paper's unsolved system was never printed. Machine results to date (EXP-050 through EXP-055): sampled shapes at the original degrees and on the reduced polygons are all partner-free (seconds per certificate); on the stratum with the forced top edge $y^8(xy-t)^8$ symbolic, every cleared certificate pairs to the CONSTANT 576 (all t, identical across sampled lower strata; five-row support); rigid universality of that covector was REFUTED by the machine (51 entering lower combinations), which pointed to the perturbative construction $\Lambda(\varepsilon)$; its base is sound (the right-hand side is the constant x^2 vector; Λ_0 globally left-null, symbolic t) and FIRST-ORDER UNIVERSALITY is proved: all 26 entering lower monomials admit correctors with the 576 pairing pinned and localized supports. Order-1 termination fails (measured), so the corrector ladder continues at order 2; each rung is a batched linear solve, and termination at any finite order yields the explicit universal covector, closing the reduced stratum for all parameters. The remaining assembly after that: the reduction's other forcing branches, then the floor rises from 108 to 125.

The closure of the $(72, 108)$ case (validated 2026-07-22)

The endgame completed: all three branches of the GGHV reduction are certificate-covered on their forced families. Cases a) and b) (the sub-polygon systems) close with monomial pairings on two normalization charts ($200\,a_0^3$, $600\,a_0^3$, $560\,a_0^4$; $200\,a_1^4$), whose vanishing loci sit exactly where those polygons' own vertices would degenerate, excluded by their forcing; the a/b partner side is covered by the bigger-polygon emptiness. Case c) closes under the torus gauge (the forced-edge parameter normalizes to $t = 1$; verified concretely by transporting an orbit sample with identical verdict): the β-symbolic certificates give pairing $23592960\,\beta$ with the degenerate $\beta = 0$ corner covered by the stratum certificates, and the interior coefficients certify axis-symbolically with constant pairings (368640 at every tested point), on top of dense mixed-direction sampling. The orientation swap is absorbed by $Q \to -Q$. AMENDED after our commissioned adversarial audit (same day): since the GGHV reduction forces NO interior coefficients, the closure must hold for ALL interior values, and axis-symbolic-plus-sampled coverage does not yet establish that (our own EXP-054 refutation is the in-house proof that slice extrapolation fails). The certified statement is therefore: **the three reduced branches are certificate-covered on their forced-edge families, with interior generality axis-symbolic and densely sampled: strong evidence toward, not yet a proof of, the $(72, 108)$ closure**; the floor-raise to 125 becomes assertable only after the simultaneous-symbolic interior certificate (the ranked hardening queue is in `program/jacobian-conjecture/72108-closure-statement.md`).

The truncation ladder for the simultaneous certificate (2026-07-24)

The certified statement above reduces the floor-raise to one object: a covector $\Lambda(\varepsilon)$, polynomial in the interior coefficients ε, that closes the reduced system for *all* interior values at once. We report the machine campaign that decides this object degree by degree; every step is exact, and every modular reduction of a rational uses $\mathrm{num} \cdot \mathrm{den}^{-1} \bmod p$ (a bug that truncated instead, see below, was caught by a declared regression gate).

Solvability is automatic (the dual annihilation identity, EXP-068). The reduced system M_0 has corank one; its cokernel covector Λ_0 satisfies $\Lambda_0 M_0 = 0$ and, more, it annihilates every Jacobian-bracket image that arises in the corrector recursion. Hence the consistency (solvability) conditions of the ladder hold at every order, and the only obstruction to a finite covector is the top-order *vanishing* condition. This is the dual form of the annihilation lemma $J(m, P^k) = -k\,L(P^{k-1}m)$ of Section 4.

The tower, decided through degree two. A polynomial covector of ε-degree d exists iff a finite linear system over the interior coefficients is feasible. We find [MV]: **no covector of degree** 1 (EXP-067; eight "diagonal" operators obstruct, the full system infeasible modulo two independent primes, hence over $\mathbb{Q}$); and **no covector of degree** 2 (EXP-072; the support $\{(0,1),(1,0),(3,5)\}$ obstructs, again conclusive modulo two primes). At degree 3 no obstruction is found through the pair and all triple supports (EXP-071, EXP-073); the quadruple tier and a constructive solve are in progress, subject to the one-sidedness noted below. The obstruction *persists* rather than migrating: the degree-two obstructing support $\{(0,1),(1,0),(3,5)\}$ contains two of the eight operators that already obstruct at degree one, and what grows is the support size needed to expose it (single operators at degree one, a triple at degree two). An earlier revision of this manuscript reported instead that the obstruction *moves* to an unrelated mixed pair; that reading rested on the retracted computation described next and is corrected here.

An honesty note (EXP-070, retracted). A first "degree 2 closed" result was withdrawn: its modular assembly reduced rationals by integer truncation, and a regression gate declared before the next experiment refused to reproduce a known exact decision, exposing the bug. Degree two was then re-decided soundly at a different support. We keep the retraction in the record; the declared-gate discipline is part of the method.

One structural reduction, and one claim withdrawn. The rational-covector relaxation adds nothing (EXP-074): since $\Lambda_0 M_0 = 0$ the certificate condition is homogeneous, so any rational covector clears its denominator to a polynomial one. A previous revision of this manuscript (v0.09, v0.10) asserted in addition that the tower has a *computable finite ceiling*, namely the dimension of the Krylov closure of Λ_0, and concluded that the campaign is a finite decision procedure. **That assertion is withdrawn here** (EXP-078): its derivation presupposes that the pinned corrector ladder terminates, and EXP-064 measured that it does *not* (the descending chain stabilizes nonzero at dimension 39). Since the pinning carries a large gauge freedom, neither the ceiling nor its negation follows; we currently have *no* a priori degree bound, effective or otherwise, and the chain of solvable degrees is controlled only by Noetherianity, which is not effective.

What survives is weaker but exact. The set of degrees at which a covector exists is *upward closed* (a degree-d covector is a degree-$(d+1)$ covector with vanishing top coefficients), so either none exists at any degree, or there is a minimal d_0; with degrees 1 and 2 closed, $d_0 \geq 3$. And the support-restricted sweeps are *one-sided*: they test necessary conditions, so an infeasible support decides a degree negatively, while all supports passing does *not* establish existence. Settling degree three positively therefore requires constructing a covector, not sweeping. Both readings of the outcome remain open and we state them plainly: a covector would be an exclusion proof, and a proof that none exists at any degree would be the first step of a counterexample.

A lead toward a degree-uniform statement. Written as a connection, the annihilation lemma makes the covector obstruction a first cohomology class, and termination of the corrector ladder becomes the regularity type of that connection (regular singular: the ladder terminates and a structural degree bound follows; irregular: no such bound, matching the measured nontermination of the pinned ladder). Whether this decides $(72, 108)$ in one stroke is the current deep question; it is stated as a program item, not a result.

Original-pair source gates for the open chain (EXP-095/096; validated 2026-07-25)

The reduced Proposition 4.3 system is a Laurent pair with bracket x^2, so theorems requiring a polynomial Keller pair with bracket one cannot be applied to it directly. Returning instead to the hypothetical original pair gives an exact applicability bridge [MV]. Its degree-72 component has

$$2A_0 = (16, 56), \qquad 2A_0' = (2, 0), \qquad 2A_1 = (11/2, 14),$$

which is precisely the first retained $D = 72$ branch in the Newton-resolution list of Makar-Limanov–Trakhtenberg [13]. Thus that independent classification retains the GGHV branch; it does not exclude it.

Two sharper necessary conditions do result. First, instantiating Lee–Li's inner-polynomial region [12] with

$$(a, b, m, n) = (2, 3, 16, 56), \qquad \mathfrak{m} = \frac{139}{39},$$

gives [MV] exactly seven possible northeastern vertices whenever the inner or innermost polynomial is nonzero:

$$(1, 3), (2, 7), (3, 10), (4, 14), (5, 17), (6, 21), (7, 24).$$

The zero-inner-polynomial alternative remains. Nor does the stronger diagonal corollary apply, since

$$\frac{b}{a} = \frac{3}{2} \not> \frac{n - m}{a} - 1 = 19.$$

Second, the approximate-root formulas of Guccione–Guccione–Horruitiner–Valqui [11] become an exact intersection invariant [MV]. For the complete chain

$$(m, n) = (3, 2), \qquad A_1 = (11/4, 7), \qquad k = 1, \qquad l = 4,$$

the forced fourth-power edge supplies four final major classes. Each has $|D^P_\tau| = 3 \cdot 7 = 21$ and $\lambda^Q_\tau = 1/4$, hence

$$I(P, Q) = \deg_x \operatorname{Res}_y(P, Q) = 4 \cdot 21 \cdot \frac{1}{4} = 21.$$

The degree-108 component therefore has 84 major and 24 minor roots. These are original-pair rejection gates independent of sampling the 51 interior coefficients of the reduced system. They do not prove that any of the seven vertices is realizable, transport either invariant through the Laurent normalization, close $(72, 108)$, or raise the degree floor. In particular, the approximate-root paper proves only an inequality for the minor-root expression where equality would be needed for the proposed family exclusion; that route is not used here.

The transport typing gate (EXP-097). These original-pair invariants cannot be imposed directly on the 51 final Laurent coefficients without additional boundary data. If $R(x) = \operatorname{Res}_y(F, G)$ and $\phi_c(x, y) = (x^{-1}, x^c y)$, then [D; exact symbolic controls]

$$\operatorname{Res}_y(\phi_c F, \phi_c G) = x^{cpq} R(x^{-1}),$$

where $p = \deg_y F$ and $q = \deg_y G$. For the final polygons $(p, q, c) = (16, 24, 4)$, so $cpq = 1536$: an input exponent interval $[\nu, 21]$ becomes $[1515, 1536-\nu]$, of width $21-\nu$. The reduction first swaps the selected coordinate eliminant and then localizes x, making x^ν a unit; Proposition 4.3 does not retain the missing $x = 0$ divisor order ν. Thus absolute degree 21 remains a reconstruction gate for an original polynomial pair, not a direct equation on the reduced coefficients. Laurent exponent width is a conditional replacement only after proving $\nu = 0$ and swap compatibility. A complete boundary-divisor ledger could supply that missing theorem; none is claimed here.

Constructible certificate strata and the first exact chart transition

The earlier phrase "finite chart cover by localized certificates" needs a commutative-algebra qualification (EXP-098). Suppose principal opens $D(s_i)$ cover the parameter space and each carries a localized lift c_i with

$$c_i^T M = 0, \qquad c_i^T b = 1.$$

Clearing denominators gives global syzygies pairing to powers $s_i^{N_i}$. Those powers generate the unit ideal, so a polynomial combination is one global covector pairing to 1. A pure principal-open cover of lifted syzygies is therefore not stronger than the global-covector target.

The strict generalization is a *constructible rank stratification*: after certifying a generic open, quotient by the residual closed-locus ideal and recompute the kernel. New syzygies can appear after this non-flat specialization and need not lift. EXP-098 verifies the distinction on an exact universally inconsistent control whose global pairing ideal is the proper ideal (x), while a new certificate appears on $V(x)$.

This mechanism now reaches the actual GGHV matrix (EXP-099–101). Common-flag shortcuts fail, but they isolate the first genuine interior interaction:

$$s = \varepsilon_{(0,1)}, \qquad t = \varepsilon_{(1,7)}.$$

For the selected 125×125 augmented minor, a Sylvester reduction gives the exact factorization [MV]

$$\frac{\det A(s,t)}{\det A_0} = \frac{(st-8)^6\big(2^{15}s^9 - (st-8)^7\big)}{2^{39}}.$$

The mixed coefficient also follows independently from

$$\operatorname{tr} B_s = \operatorname{tr} B_t = 0, \qquad \operatorname{tr}(B_s B_t) = \frac{13}{8}.$$

At $(s,t) = (-8,-1)$, the first minor vanishes, but exact ranks remain

$$\operatorname{rank} M = 124, \qquad \operatorname{rank}[M \mid b] = 125.$$

Pivot extraction gives an alternative right-hand-side-containing minor nonzero there. The first two minors cover the component $st = 8$ and leave only

$$2^{15}s^9 = (st-8)^7,$$

with normalization

$$s = 8u^7, \qquad t = \frac{8u^9+1}{u^7}, \qquad u \neq 0.$$

At $u = 1$, both old minors vanish but a third exact augmented minor is nonzero (EXP-102). Its dense pullback has combined direction rank 121 and hit the declared five-minute budget. EXP-103 replaces that backend by root-of-unity evaluation and NTT interpolation of complete maximal-minor polynomials. An endpoint gate catches an 81-degree cancellation that a modular gcd alone would miss.

EXP-104 resolves that gate over characteristic zero [MV]. The third determinant is evaluated exactly at 100 integer values. Its assignment bounds are $[1547, 1646]$; after division by u^{1547}, exact degree-99 interpolation shows that coefficients $0, \ldots, 80$ vanish and coefficient 81 does not. Thus its exact support is $[1628, 1646]$. A second row chart has exact support $[777, 903]$.

The support graphs of both charts admit connected row/column weights modulo 9 (EXP-105). With $z = u^9$, removal of the proved Laurent monomials and integer contents gives the primitive determinants

$$F(z) = 21 - 96z - 1024z^2, \qquad G(z) = (8z + 1)^{14}.$$

The only zero of G is $z = -1/8$, where $F(-1/8) = 17$. More strongly, the artifact contains integer polynomials A, B satisfying

$$A(z)F(z) + B(z)G(z) = 17^{14}.$$

Therefore the two normalized maximal minors generate the unit ideal over $\mathbb{Q}[u, u^{-1}]$. Together with the first two charts, this proves rank 125 for the augmented matrix at every point of the declared $\{(0, 1), (1, 7)\}$ coefficient slice [MV].

The grading is not confined to that slice. For every nonconstant lower monomial $x^p y^q$ tested in the persisted 26-parameter family, EXP-106 finds the exact variable-weight law

$$w_{p,q} \equiv q - p + 1 \pmod 9.$$

All 23 remaining nonconstant directions preserve both connected chart gradings; the constant direction is bracket-zero. This supplies a scalable next coordinate system, not a full-family proof. The promoted direction $(0, 7)$ has selected ranks 53 and 41; after setting $z = u^9$ and $y = v/u$, both determinant charts have z-width 14.

EXP-107 reconstructs both bivariate minors on independent 16-by-64 NTT grids and verifies them by off-grid determinant evaluations [MV]. The endpoint-safe minor is independent of y, so it remains $G(z) = (8z + 1)^{14}$. The other minor restricts at $z = -1/8$ to a squarefree degree-12 polynomial $Q(y)$. Thus the first two charts leave only twelve geometric points in that fiber, not a curve.

EXP-108 selects a third 125-row chart at the residual fiber [MV]. Removing the connected mod-9 row/column weights turns both fiber matrices into exact rational matrices. Interpolation under structural degree bounds 14 and 13, using 29 exact determinants and two independent controls, gives primitive polynomials $Q, H \in \mathbb{Z}[y]$. The polynomial Q is irreducible over $\mathbb{Q}$, while, up to sign,

$$\begin{aligned} H(y) = y(49y^2 + 168y + 192)&(2401y^4 + 1568y^2 + 1024) \\ &\cdot (117649y^6 - 403368y^5 + 921984y^4 - 1053696y^3 + 786432). \end{aligned}$$

Their exact gcd is one. The persisted integer Bézout identity has nonzero constant

$$303627789524487856472770343437884885512193900741932869903974 4.$$

Since every common zero with G must lie over $z = -1/8$, the three minors prove rank 125 throughout the declared $\{(0, 1), (1, 7), (0, 7)\}$ coefficient slice.

These results establish the first complete constructible chart cover on an actual three-coefficient GGHV slice and expose a global lower-family grading. They do not cover simultaneous variation of the other coefficients, exclude the full $(72, 108)$ family, or raise the degree floor.

Full-family rank audit and the exact 36-column core

The next full-family audit corrects a tempting but vacuous rank target. The Q-polygon contains the constant monomial, whose column is identically zero:

$$[P, 1] = 0.$$

Hence $\operatorname{rank} M(\varepsilon) \leq 124$ for every parameter value. The exact nonzero pinned augmented minor from EXP-059 then proves that $\operatorname{rank} M = 124$ and $\operatorname{rank}[M \mid b] = 125$ generically [D]. This is inconsistency on a nonempty Zariski-open subset, not on every fiber. The unresolved object is the common zero locus of augmented maximal minors. EXP-111 also corrects the row pool: the forced-only construction has 289 rows, while the complete union inside the same reduced pool has 302 rows [MV].

EXP-112 removes the structural constant column, retains the target column, and normalizes all 51 exact perturbation matrices on a pinned 125-row chart [MV]. The union dependency graph has one 36-vertex cyclic component, three cyclic singletons, and 86 acyclic singletons. Consequently the selected determinant factors as

$$\det A_{\mathrm{sel}}(\varepsilon) = \det(A_0)(1 + \varepsilon_{(1,0)})^3 \det C_{36}(\varepsilon),$$

where C_{36} depends on only 24 parameters. The other 27 directions are determinant-inert on this chart. On the forced axis,

$$\chi_{N_{(1,0)}|C_{36}}(\lambda) = \lambda^{23}(\lambda - 1)^{13},$$

recovering the total factor $(1 + \varepsilon_{(1,0)})^{16}$.

Graph support is not determinant support. EXP-113 finds two deletion-minimal triples that each make the 36-core strongly connected:

$$T_A = \{(0, 1), (0, 7), (2, 9)\}, \qquad T_B = \{(0, 1), (0, 5), (1, 0)\}.$$

EXP-114 computes their exact determinants [MV]. On T_A, the graph-active $(2, 9)$ direction cancels identically, and the degree-21 determinant has only 24 monomials and factors in degrees $3, 12, 6$.

On T_B, write

$$a = \varepsilon_{(0,1)}, \qquad b = \varepsilon_{(0,5)}, \qquad d = 1 + \varepsilon_{(1,0)}.$$

Then

$$\det C_{36}|_{T_B} = 2^{-42} G_{54}(a, b, d) H_{63}(a, b, d),$$

where

$$\begin{aligned} G_{54} = {} & 30720000a^6b^4 + 48828125a^3b^{11} + 150000000a^3b^8d \\ & + 64000000a^3b^5d^2 - 39321600a^3b^2d^3 + 16777216d^6, \end{aligned}$$

and

$$\begin{aligned} H_{63} = {} & 4096a^9 + 184320a^6bd^2 - 1800000a^3b^5d^3 + 1843200a^3b^2d^4 \\ & + 1953125b^9d^4 + 3000000b^6d^5 + 1536000b^3d^6 + 262144d^7. \end{aligned}$$

These factors are weighted homogeneous for $\mathrm{wt}(a, b, d) = (7, 3, 9)$, of weighted degrees 54 and 63. They expose a weighted open/boundary split for the next chart calculation: normalize $d \neq 0$, then treat $d = 0$ with alternative minors from the complete 302-row system. No result in this subsection excludes the residual factor loci, the full $(72, 108)$ family, or $JC(2)$.

Weighted residual transitions and the $d = 0$ quotient

EXP-115 first normalizes the open chart $d \neq 0$ by

$$a = Au^7, \qquad b = Bu^3, \qquad d = u^9.$$

Writing $X = A^3$, $G_{54}(A, B, 1)$ remains irreducible, while

$$H_{63}(A, B, 1) = L(A, B)Q(A, B)$$

splits into irreducible degree-three and degree-six factors over $\mathbb{Q}[A, B]$ [MV]. Exact good-prime alternative minors prove that none of G_{54}, L, Q is wholly contained in the rank-deficient locus. The L-transition also lifts to the rational point $(A, B, d) = (0, -4/5, 1)$. Proper intersections on these components remain.

The boundary $d = 0$ is structural. Cancelling the forced x-term puts

$$P = ay + by^5 + y^8(1 - xy)^8$$

inside the retained Q-space, so $[P, P] = 0$ is an exact polynomial right kernel [D]. Its y^8-coordinate is the unit 1; deleting that column gives a 302-by-124 quotient augmented matrix with 123 coefficient columns. The origin has exact rank profile 112/113, while the nonzero controls $(0, 1)$, $(1, 0)$, and $(-9, 12/5)$ have profile 123/124 [MV].

EXP-116 compresses the quotient graph into cyclic blocks of sizes

$$51, 11, 10, 9, 8, 7$$

and sixteen cyclic singletons. EXP-117 computes the separately budgeted 51-block and all smaller determinants. In original coordinates the selected quotient determinant is, up to a persisted nonzero rational scalar,

$$\det A_{\text{quot}} \doteq b^{32} F_{28}(a, b) \prod_{c \in \mathcal{C}} (ca^3 + 78125b^7), \qquad \mathcal{C} = \{49152, 32768, 16384, 8192, 4096\},$$

where

$$\begin{aligned} F_{28} = {} & 13750615562268966912a^{12} - 38922964690796544000000a^9b^7 \\ & + 614959441120000000000000a^6b^{14} \\ & + 8653485107421875000000000a^3b^{21} \\ & - 711902976036071777343750b^{28}. \end{aligned}$$

All factors are homogeneous for $\mathrm{wt}(a, b) = (7, 3)$. On $b \neq 0$, set $z = a^3/b^7$. The residual becomes one squarefree degree-nine equation: five rational roots

$$-\frac{78125}{4096}, \quad -\frac{78125}{8192}, \quad -\frac{78125}{16384}, \quad -\frac{78125}{32768}, \quad -\frac{78125}{49152},$$

and four roots of the quartic obtained from F_{28}/b^{28} [MV]. Thus the selected quotient residual is reduced to $b = 0$ and nine invariant points.

EXP-118 supplies the exact alternative chart [MV]. Its complete quotient determinant is

$$\det A_{\text{alt}} \doteq a^{107}.$$

The persisted scalar is a nonzero integer. This chart covers $a \neq 0$, including all nine invariant points. On $a = 0$, the selected determinant is exactly b^{95}, and at the origin the quotient rank profile is 112/113. These three cases close the complete $d = 0$ quotient plane. This is a boundary-stratum closure, not a result about $d \neq 0$.

Exact weighted-open charts and component ideals

EXP-119 verifies the full 302-by-125 augmented matrix is covariant for $\mathrm{wt}(a,b,d) = (7,3,9)$ [MV]. On $d = 1$, a first alternative row basis gives an exact determinant with 114 monomials. Writing $X = A^3$, it factors as a nonzero rational scalar times

$$B^{36}K(X,B) \prod_{c\in\{4096,8192,16384,32768,49152,65536,86016\}} (cX + 78125B^7),$$

where K has degree 6 in X, degree 22 in B, and 30 monomials. Its gcd with each irreducible selected component G, L, Q is one. Nevertheless, the three exact elimination resultants are nonconstant, of degrees 852, 324, and 648 in B. Hence this chart removes every curve component but leaves three finite proper intersections.

EXP-120 evaluates the independently selected G-basis [MV]. The first rational normalization $(A,B,d) = (1,0,1)$ works, the largest cyclic block has size 25, and the exact determinant has 21 monomials. Up to a nonzero rational scalar its invariant form is

$$\Delta_G = X^{30}L(X,B)Q(X,B)R(X,B),$$

where

$$R = -156250B^7 + 15625B^6X + 28000B^3X + 3200B^2X^2 + 1024X.$$

Factorwise exact Groebner calculations give

$$(G, \Delta_{LQ}, L) = (G, \Delta_{LQ}, X) = (G, \Delta_{LQ}, Q) = (G, \Delta_{LQ}, R) = (1).$$

Therefore the three charts cover the complete G component [D].

The corresponding ideals on L and Q are nonunit but zero-dimensional. The L lexicographic eliminant has degree 108 in B and squarefree degree 73. For Q, a six-element graded-lex basis proves zero-dimensionality; the optional FGLM conversion reached its declared 180-second gate, so no lex eliminant or point count is claimed. The explicit LQ factor in Δ_G shows that this basis cannot remove either remaining component. The next exact charts must be selected directly on those finite residual schemes.

Together, EXP-118–120 close the $d = 0$ quotient boundary and the G open component, but not the L/Q finite residuals. They do not close the full T_B restriction, the 24-parameter core, the full 51-parameter family, $(72,108)$, the planar degree floor, or $JC(2)$.

The frontier, corrected. Beyond 125, our first probe case (the chain $(8,40)/(8,28)/(11/4,7)$, max degree 144) turns out already excluded in the published literature: the closing remark of [8] states that the corner data $(8,28),(8,40)$ forces $A'_0 = (8,4)$, which is not a possible last lower corner. We had not fully mined that remark; several of the twenty-four configurations in $[125,150]$ may be similarly settled, and a re-audit against the excluded family $\wp(n', n'-1)$ precedes any new attack (EXP-082, EXP-083).

The counterexample as an incidence map, and its exclusion in the plane

A complementary lens sharpens why the two-variable case resists the very mechanism that settled dimension three. The 2026 counterexample is not an arbitrary polynomial map: it is the coefficient map of the product of a linear and a quadratic binary form, restricted to a normalized slice. Concretely, on $V_1 \times V_2 \to V_3 \times \mathbb{A}^1$, $(L,Q) \mapsto (LQ, \mathrm{Res}(L,Q))$, the slice

$X_H = \{(L, Q) : \mathrm{Res}(L, Q) = 1,\ [LQ]_{\mathrm{next}} = 1\}$ carries the projection $(L, Q) \mapsto LQ$, which is étale on the coprime (nonzero-resultant) locus by a one-line argument: multiplication forgets exactly the relative scaling $(L, Q) \mapsto (\lambda L, \lambda^{-1} Q)$, and the resultant, of weight $\deg Q - \deg L$ under that scaling, detects it. The projection is generically n-to-one (the choices of linear factor), hence not injective. We verified [MV] the full identification: the displayed counterexample equals this coefficient map up to linear normalization, with $X_H \cong \mathbb{A}^3$. So, for this family, the entire Jacobian question reduces to a single *recognition of affine space*: is $X_H \cong \mathbb{A}^n$?

In dimension three the recognition holds by a toric coincidence: the residual $\mathbb{G}_m$-grading has weights $(1, -1, -2)$, and the associated semigroup generators form a unimodular lattice basis, so $X_3 \cong \mathbb{A}^3$. In the plane it *fails*, and we can say why in one invariant. The controlling quantity is $n-2 = \deg Q - \deg L$, the resultant weight under the relative scaling; it is simultaneously the degree of the Danielewski core $\{x^{n-2}W = B^{n-2} - 1\}$ and the number of boundary components. For $n = 2$ it is 0: the resultant becomes scaling-invariant, unable to rigidify the normalization, and the slice degenerates (reducible; equivalently, the finite-root model is $\mathbb{A}^2$ minus the discriminant, which carries a nonconstant unit and is therefore not $\mathbb{A}^2$). For $n = 3$ it is 1, the unique value that is both nonzero and linear, giving $X_n \cong \mathbb{A}^n$; for $n \geq 4$ it is ≥ 2, and the class group obstructs. Thus the exact structure of the known counterexample cannot occur in the plane [MV], a conclusion consistent with the companion's $\mathbb{G}_m$-equivariant rigidity theorem, which forbids the torus-graded planar analog independently. This excludes the incidence/root-selection class, the structure of every constant-Jacobian noninvertible map known to date; it is strong structural evidence for $JC(2)$, not a proof, since a planar counterexample of some entirely different structure is not addressed.

6 Positioning and novelty

Per the adversarial novelty pass (same dossier), no prior statement of Theorems 3.1 or 3.2 was found. The novelty record for the companion's equivariant rigidity theorem is updated: Shaska [18], submitted on 2026-07-22 after EXP-010 was declared, independently proves the full all-signature classification. Same-sign actions may give nonlinear triangular automorphisms; opposite-sign actions give the linear case of EXP-010. No priority or novelty claim is made for that classification. Theorem 3.3 was not found as stated, but its operator identity is within the standard bracket calculus [14, 8]: the statement appears new, the method is classical. Theorem 3.4's counterexample half is essentially contained in [8] (Prop. 4.1) with van den Essen's subrectangular normalization [15]; we claim only the sharp dichotomy form. All proofs are elementary once seen; a folklore risk remains and is recorded, with the primary-source audit trail (including its open verification items) persisted in the repository.

7 Program

Priorities, from the standing routes evaluation: (1) select new maximal minor row bases directly on the finite L and Q common residuals left by EXP-120, and test the resulting ideals exactly; (2) retain EXP-118's complete $d = 0$ cover and EXP-120's G-component cover as regression gates rather than recomputing them; (3) retain the exact EXP-108 lift through $(0, 6)$ as a bounded chart control rather than a terminating programme; (4) reopen the intersection-21 transport only after constructing a complete boundary-divisor ledger through the swap, Laurent cuts, and final inversion; (5) pursue Newton resolution only after deriving a condition beyond the already-retained first $D = 72$ branch; and (6) retain the properness reformulation ($JC(2)$ holds iff every planar Keller map has empty Jelonek set; exact certificates shipped, EXP-014) as a cross-check instrument. The reproducibility contract: every [MV] claim has a persisted script and output; hypotheses are declared before runs; refuted predictions remain in the record.

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